# Subset antichains mathematical overview

Adult theorem map | Draft and unpiloted

## Objects and order

Each card is a subset of a fixed set of n distinct symbols, with each symbol present at most once. The full deck has all 2 to the power n subsets, including the empty card. A fits inside B when every symbol of A occurs in B. A legal collection is an antichain: no two different selected cards contain one another. Sharing some symbols is allowed; duplicate cards are not.

## The certificate used here

A nested row is a chain under inclusion. If every card is put in exactly one of r chains, any antichain has at most r cards, because it takes at most one from each chain. An antichain of r cards together with an r-chain partition proves both the largest collection size and the fewest rows. No search through all possible collections is needed. Each full deck has such a cover with exactly as many rows as its largest collection (for any finite deck this is Dilworth's theorem), so the certificate always exists.

## Exact maxima and broader theorem

For the full two-, three- and four-symbol decks the maxima are 2, 3 and 6; the guide gives matching partitions. On four symbols the unique six-card antichain is the family of all two-symbol cards. More generally, Sperner's theorem gives maximum size binomial(n, floor(n/2)), attained by all subsets of a middle size. The packets establish the small cases with explicit certificates; they do not prove the general theorem. An unextendable (maximal) collection need not be largest (maximum).

## Across the bands

K-1 compares picture cards, collects small families and tries nested rows. Grades 2-3 organizes all small choices and uses a chain partition as an upper bound. Grades 4-5 distinguishes maximal from maximum, proves the four-symbol optimum, investigates a forced card and classifies the largest collections.

## Discovery and proof

Build examples first, conjecture a best size, and then check that a chain arrangement covers every card exactly once. Failed attempts to add another card prove only that the current collection cannot be extended. Matching constructions and universal bounds explain the optimum; the detailed partitions and completeness arguments follow.

# No card inside another facilitator guide

**Week 19 is an unscheduled library label. Draft and unpiloted.** Packets: K-1 (F19-K-v3, 5 pages), Grades 2-3 (F19-23-v3, 6 pages) and Grades 4-5 (F19-45-v3, 6 pages), each with eight problems, and the adults' card sheet (F19-C-v1). The mathematical goal is to build a large legal collection and justify that no larger one can exist.

## Supplies and preparation

For each of three tables: two complete 16-card decks, blank paper, pencils, 20 selection counters, and an uncommitted supply of blank paper strips. Total: six decks, 96 cards, and 60 counters. Print the card sheet, week-19-materials.pdf (sixteen 4.5 cm cards with circle A, triangle B, square C and star D in fixed corners), six times at 100%, each copy on a different colour of paper, so every deck has its own colour and duplicate copies never enter one active deck. Every card, the empty card included, carries the same short mark on its top edge: it shows which way is up and is not a picture.

The student sheets show small reference pictures; children handle the cut cards. Allow about 20 minutes to cut the six sheets with a paper cutter, stacked, or about an hour with scissors (estimates; the cutting is untested). Adults cut before the session and check each deck against the inventory below. For a two- or three-symbol deck, retain only cards avoiding unused symbols; do not erase symbols and create duplicate cards.

| Number of symbols on a card | Cards in a four-symbol deck |
|---|---|
| 0 | empty |
| 1 | A, B, C, D |
| 2 | AB, AC, AD, BC, BD, CD |
| 3 | ABC, ABD, ACD, BCD |
| 4 | ABCD |

## Shared launch and flexible hour

Give 3 minutes to handle cards. In about 3 more minutes, show that A fits inside AB by pointing to the circle on both, that AB and AC may be together although they overlap, and that empty fits inside every card. Then build one collection that is in no task: BC and BD (they share the triangle; neither fits inside the other); try B and take it back after pointing to its triangle on BC; add ACD. Sketch it on the board as three boxes with the pictures in their corners. Collections and rows stay on the table to be checked; children sketch only finished arrangements. Do not display a chain partition or announce the target size. Checking is the partner's job: when a card is placed, the partner points to every picture of the smaller card on the larger one. Partners swap with each card; at the 4-5 table the third child referees. The adult steps in when partners disagree.

After the run and the launch there are about 35-40 working minutes: about 15 for small collections (K-1 P1-P4; middle P1-P3; upper P1-P3), 15 for further search and nested rows, and 5-10 for a bound or an extension (K-1 P7-P8; middle P7-P8; upper P6-P8); then share one construction and one reason no larger choice works. These are choices, not targets.

# Readiness and the small-deck certificates

## Prerequisites and gates

K-1 entry: match picture symbols, compare two cards by inclusion, count up to three; an adult reads and records. Middle entry: organize an exhaustive small list and sort every card into one row. Upper entry: distinguish an unextendable collection from a genuinely largest collection and connect a covering to a universal upper bound. No factorials, binomial coefficients, or formal set notation are required.

**Gate before maximizing:** children can explain why A/AB is forbidden but AB/AC is allowed. **Gate before the certificate:** they can make a valid nested row and verify that every active card appears once. If they are not ready, return to two cards and point to individual pictures rather than describing a rule abstractly.

## Guide notation

A is a circle, B a triangle, C a square, D a star. AB means one card with both symbols. A/B means a collection of two separate cards. The word empty names the blank card; it is different from choosing no cards. The empty card is comparable with everything. Two different cards of the same size cannot contain each other.

## The two-symbol deck

| Nested row | Legal collection meeting each row |
|---|---|
| empty -> A -> AB | A |
| B | B |

The collection A/B has size two. The displayed rows use each of the four cards exactly once. A legal collection takes at most one card from each row because any two cards in the same row are comparable. Thus no collection has more than two, and two rows is also the minimum possible row count.

## The three-symbol deck

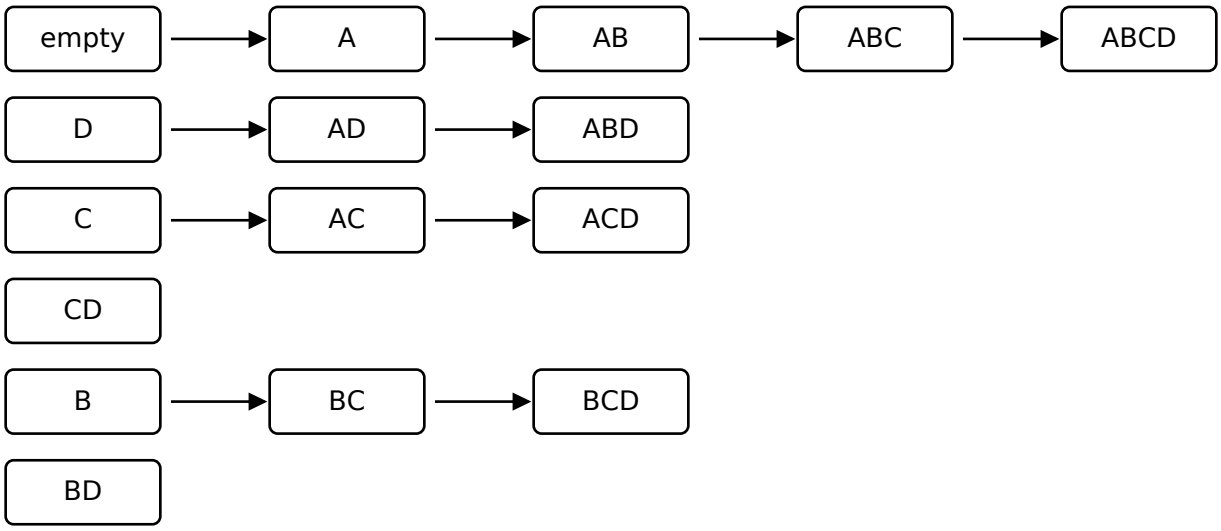
| Nested row | One chosen card |
|---|---|
| empty -> A -> AB -> ABC | A |
| C -> AC | C |
| B -> BC | B |

These three rows cover the eight cards exactly once. The legal collection A/B/C supplies three cards, so the maximum collection size and minimum row count are both three. AB/AC/BC is another largest collection. This is a proof covering every possible selection, not a record of several unsuccessful attempts.

**Keep the method available, not compulsory.** Let children rearrange cards freely. Introduce the one-per-row question only after they have a nested arrangement or need a way to justify their best collection. Do not preallocate exactly the optimum number of row strips.

# A complete four-symbol certificate

Keep this adult answer diagram hidden until it is useful. Many valid six-row partitions exist. The following one covers all sixteen cards exactly once; each arrow adds symbols and never removes any.




Every arrow means strict containment. Every one of the 16 cards appears once.

## Why the optimum is exactly six

The six two-symbol cards AB, AC, AD, BC, BD, CD form a legal collection: distinct equal-sized cards cannot contain one another. This proves that six can be achieved. The diagram partitions the entire deck into six nested rows. Any legal collection meets each row at most once; therefore no legal collection can exceed six.

The same two observations prove the row count is minimal. Any partition into nested rows must put the six two-symbol cards into six different rows, since no pair of them is comparable. Six rows are both necessary and sufficient. A different arrangement is valid if every card appears exactly once and every row is nested.

## How to verify a child's certificate

First check coverage against the 16-card inventory: no missing or repeated card. Then check each adjacent pair in each row. Adjacent containment implies containment for any more distant pair by following the symbols through the intervening cards. Finally ask why one legal collection cannot use two cards from one row. Missing even one card breaks the upper-bound certificate.

## Why an unextendable collection need not be largest

The one-card collection ABCD cannot accept any other card, but it has size one. A/BCD is also unextendable, with size two. Each is smaller than the six-card collection. Say largest for the actual optimum; reserve maximal, if used at all, for merely unextendable. Keeping these meanings separate is part of the mathematics.

# K-1 solutions for Problems 1 to 4

## Problem 1 Four small groups Student page 1

**Answer and reasoning.** In printed order, choose A/B (size 2); A/B (size 2); AB/AC/BC (size 3); A/BC (size 2). These are the unique largest choices in their respective pictured groups. In the first group empty cannot be paired with any card. In the second AB contains both A and B, while A and B do not contain each other. In the third, every card has two symbols and the cards are distinct, so none contains another despite their overlap. In the fourth, ABC contains both other cards; A and BC are incomparable.

**Hold back, then offer.** Ask the child to point to each symbol rather than deciding by overlap alone. For AB/AC, where would B fit on AC? Where would C fit on AB?

## Problem 2 Every nonempty collection from four cards Student page 2

**Answer and reasoning.** There are five: empty alone; A alone; B alone; AB alone; and A/B. Any one card is legal. Empty and AB are each comparable with every other card, so neither can appear in a larger collection. Of the remaining cards, A and B may be paired. No three-card collection is possible. The zero-card choice is excluded by the question's one-or-more condition.

**Hold back, then offer.** If a child forgets single-card collections, ask whether one card breaks the rule. Distinguish the empty card from choosing nothing.

## Problem 3 Two-card collections from eight cards Student page 2

**Answer and reasoning.** The full list of nine is A/B, A/C, B/C; AB/AC, AB/BC, AC/BC; A/BC, B/AC, C/AB. Empty and ABC cannot join another card. Among the other six cards, two distinct singletons always work, as do two distinct pairs. A singleton can accompany a pair only when that symbol is absent from the pair, giving the last three choices. The task asks for as many as the child can find; do not turn nine into a completion quota.

**Hold back, then offer.** Ask whether the cards may share a symbol. If finding new pairs becomes random, sort existing answers by the sizes of their cards.

## Problem 4 Two largest collections Student page 3

**Answer and reasoning.** The maximum is three; use A/B/C and AB/AC/BC. The three-row partition on guide page 3 excludes four. To see these are the only size-three choices: if a singleton A is chosen, only B, C, BC remain eligible. To add two compatible cards one must choose B and C. The same argument applies to any singleton. Without singletons, a three-card collection must use all three pair cards; empty and ABC cannot participate.

**Hold back, then offer.** Let children find two different triples before offering any row argument. Ask how they could be sure a fourth card cannot be added or obtained by a different start.

# K-1 solutions for Problems 5 to 8

## Problem 5 Keep the starting cards Student page 3

**Answer and reasoning.** The four starts are ABC; A; AB; A/BC. Their largest final sizes are respectively 1, 3, 3, 2. Add nothing to ABC; add B and C to A; add AC and BC to AB; add nothing to A/BC. ABC contains every card. A/B/C and AB/AC/BC attain the known maximum of three. For A/BC, every other card is comparable with A or BC: B and C fit in BC; AB and AC contain A; empty fits in either; ABC contains both. Thus its two-card collection is unextendable but not largest.

**Hold back, then offer.** Ask whether the starting cards may be replaced; here they must stay. A collection that stops at two does not contradict the existence of a different three-card collection.

## Problem 6 Fewest nested rows Student page 4

**Answer and reasoning.** For the four-card deck the minimum is two rows: empty -> A -> AB, and B. For the eight-card deck the minimum is three: empty -> A -> AB -> ABC; C -> AC; B -> BC. Every card appears exactly once in each arrangement. The legal singleton collections A/B and A/B/C force at least two and three rows, respectively: two incomparable singleton cards cannot be in one nested row. The displayed partitions attain those lower bounds.

**Hold back, then offer.** Check nesting locally by placing cards left to right. If too many rows are used, ask whether a loose card can join a row without disturbing its containment order. Do not hand out exactly two or three strips.

## Problem 7 Could four cards work Student page 5

**Answer and reasoning.** No. The maximum is three. Use any complete three-row partition from P6. Choosing four cards would put two in the same row, where one contains the other. The argument applies to every choice of four cards, not only a child's current attempt. Children can physically place four selection counters on the three rows to explain the conflict.

**Hold back, then offer.** Ask whether four counters can occupy three rows with no two in the same row. Then connect the two counters to the collection rule.

## Problem 8 Remove any one card Student page 5

**Answer and reasoning.** Yes, a three-card legal collection always remains. The two largest collections A/B/C and AB/AC/BC share no cards. Removing one card can damage at most one of them. If a singleton is removed use all pairs; if a pair is removed use all singletons; if empty or ABC is removed, either works. This covers all eight choices of removed card.

**Hold back, then offer.** Ask the child to keep the two triples from P4 visible and point to which one the removed card affects. The surviving triple need not be the same for every removal.

# Grades 2-3 solutions for Problems 1 to 4

## Problem 1 Every legal nonempty collection Student page 1

**Answer and reasoning.** The five collections are empty alone, A alone, B alone, AB alone, and A/B. All four single-card choices work. A collection with empty or AB cannot have another card. With those excluded, only A and B remain, and their pair works. This proves the list complete, including the impossibility of any three-card collection.

**Hold back, then offer.** Ask whether choosing the empty card is the same as choosing zero cards. Then ask which cards could possibly belong to a multi-card collection.

## Problem 2 Every largest collection from eight cards Student page 2

**Answer and reasoning.** The maximum is three, attained by exactly A/B/C and AB/AC/BC. The three-row certificate on guide page 3 proves no fourth card is possible. For completeness of the maximum list, a singleton forces the other two selected cards to be the other singletons: among B, C, BC only B/C is a compatible pair. Without a singleton, empty and ABC cannot join a triple, leaving only the three two-symbol cards.

**Hold back, then offer.** Ask whether a largest collection could mix one-symbol and two-symbol cards. Let the child analyze a fixed singleton rather than testing every three-card choice.

## Problem 3 Three required starts Student page 2

**Answer and reasoning.** For ABC, keep it alone: final size 1. For A/BC, add nothing: size 2. For AB, add AC and BC: size 3. Being unable to add another card does not imply largest size. ABC conflicts with every other card. For A/BC, every unused card contains A, fits inside BC, or is empty/full, so no addition works. The third construction attains three, proving both previous stopped collections are smaller than a maximum.

**Hold back, then offer.** Ask whether two collections can both resist additions while having different sizes. Keep the prescribed cards rather than silently replacing them.

## Problem 4 Fewest rows for both decks Student page 3

**Answer and reasoning.** The row minima are two and three, with the exact partitions displayed on guide page 3. The partitions cover each complete deck once and are nested. A/B and A/B/C force at least two and three distinct rows, respectively. Thus the attained counts are optimal. Any other complete partitions with those counts are equally valid.

**Hold back, then offer.** Ask children to audit a partner's rows for one omitted or duplicated card, then explain why fewer rows cannot hold all the mutually incomparable singleton cards.

# Grades 2-3 solutions for Problems 5 to 8

## Problem 5 Rule out every four-card collection Student page 4

**Answer and reasoning.** Four cards cannot work in the three-symbol deck. Use the complete three-chain partition from P4. If four selected cards occupied three rows, at least two would share a row and be comparable. Coverage matters: a collection might otherwise choose a card that was left outside the claimed rows. The bound of three is attained by A/B/C.

**Hold back, then offer.** If the explanation lists only failed attempts, ask how the three rows control a choice the group has never tried.

## Problem 6 Search the sixteen-card deck Student page 5

**Answer and reasoning.** A largest collection is AB/AC/AD/BC/BD/CD, size six. All six are distinct cards of equal size, so none contains another. The full six-row certificate on guide page 4 rules out seven or more. At this stage accept six as a well-supported conjecture; Problem 8 asks why, and Problems 7 and 8 supply the universal argument.

**Hold back, then offer.** Ask whether sharing one symbol is actually forbidden. If a child has stopped with ABCD or A/BCD, compare with a different start rather than suggesting the stopped collection is optimal.

## Problem 7 Fewest rows for sixteen cards Student page 6

**Answer and reasoning.** Six rows suffice; the full diagram on guide page 4 is one answer. Read its rows as empty -> A -> AB -> ABC -> ABCD; D -> AD -> ABD; C -> AC -> ACD; CD; B -> BC -> BCD; BD. Every card appears once and every row is nested. At least six rows are required because the six two-symbol cards are pairwise incomparable, so no two can share one nested row. Thus six is the exact minimum.

**Hold back, then offer.** Let students build and revise rows; do not provide the completed six-row diagram until needed. If they find five rows, audit missing cards, repeated cards, and a hidden containment error.

## Problem 8 Prove the largest size Student page 6

**Answer and reasoning.** The largest size is six. The six two-symbol cards demonstrate attainability. The complete six-row partition shows any legal collection uses at most one card from each of six rows, so no larger collection can exist. Both halves are necessary: a big collection alone does not rule out larger ones, and a six-row bound alone does not show six is attainable.

**Hold back, then offer.** Ask which part of the work shows can, and which part shows cannot do better. Have the child point to actual cards and rows while explaining.

# Grades 4-5 solutions for Problems 1 to 4

## Problem 1 Small-deck exact maxima Student page 1

**Answer and reasoning.** The two-symbol maximum is two, achieved by A/B. The three-symbol maximum is three, achieved by A/B/C or AB/AC/BC. The complete two-row and three-row partitions on guide page 3 give matching upper bounds because a legal collection uses at most one card in each row. An alternative complete case analysis is valid. The task requires a reason covering every larger choice, not simply an unextendable example.

**Hold back, then offer.** Ask what a certificate would need to account for: every card in the deck, even those absent from the proposed winning collection.

## Problem 2 Sixteen-card maximum Student page 2

**Answer and reasoning.** The exact maximum is six: all six two-symbol cards. The cards are pairwise incomparable. The complete six-row partition on guide page 4 is an upper-bound certificate. Children may first conjecture six and later obtain the proof in P5-P6; do not state that a failed search for seven already proves it.

**Hold back, then offer.** Ask whether there is a way to organize all cards so that any legal collection has a visible limit.

## Problem 3 Keep three contrasting starting collections Student page 3

**Answer and reasoning.** Start ABCD: size 1, add nothing. Start A/BCD: size 2, add nothing. Start AB/AC: size 6, add AD, BC, BD, CD. ABCD contains all cards. With A/BCD, any card containing A conflicts with A, and every card avoiding A is a subset of BCD, so it conflicts with BCD. The pair-card start extends to the six-card maximum. Thus unextendable and largest have different meanings.

**Hold back, then offer.** If the group tries to add a card to A/BCD, divide candidates into those containing A and those avoiding A. Each class has a direct obstruction.

## Problem 4 Minimum nested row counts Student page 4

**Answer and reasoning.** Two rows are optimal for the four-card deck; three rows are optimal for the eight-card deck. Exact complete arrangements appear on guide page 3. Each is a partition into chains. The two or three singleton cards are mutually incomparable, forcing that many different rows; the displayed arrangements meet the lower bounds. Rows may be singletons and need not have equal lengths.

**Hold back, then offer.** Ask a group with a proposed partition to select one incomparable card from each needed row and explain why those cards could never share a row.

# Grades 4-5 solutions for Problems 5 to 8

## Problem 5 Fewest rows for all sixteen cards Student page 5

**Answer and reasoning.** The exact minimum is six. The complete six-row arrangement on guide page 4 is one solution. Coverage and nesting are directly verified from that diagram. The six two-symbol cards require six separate rows in any partition because no two of them contain one another. Thus fewer rows are impossible even for a completely different partition.

**Hold back, then offer.** Ask children to check their row count against a known large legal collection. Every card must be used once; duplicates cannot repair a missing card.

## Problem 6 Maximum with a universal certificate Student page 5

**Answer and reasoning.** Six is attainable by AB/AC/AD/BC/BD/CD, and no larger collection works. The complete six-chain partition permits at most one card per chain. Seven chosen cards would place two in one chain, producing a containment conflict. This is the requested argument about every larger collection. A child's own different six-chain partition is just as good.

**Hold back, then offer.** Ask the child to connect the numerical upper bound to a specific forbidden pair that must arise if seven cards are selected.

## Problem 7 Largest collection containing the circle card A Student page 6

**Answer and reasoning.** The maximum is four. Examples are A/B/C/D and A/BC/BD/CD. Every other chosen card must avoid A, since any card containing A also contains the required singleton. Those other cards form an antichain in the three-symbol deck on B,C,D, whose maximum is three. Adding the required A gives at most four, achieved by either example. The empty card cannot be one of the other choices because it lies inside A.

**Hold back, then offer.** Ask which cards the required singleton immediately rules out. Then identify the smaller deck formed by the survivors.

## Problem 8 Every maximum collection Student page 6

**Answer and reasoning.** There is exactly one size-six collection: AB/AC/AD/BC/BD/CD. If a collection contains a singleton, P7 (with symbol names exchanged) bounds it by four. If it contains a three-symbol card, replace every card by its complement in ABCD. Containment reverses under complement, so incomparability is preserved and the triple becomes a singleton; again the size is at most four. Empty or ABCD forces size one. Therefore a size-six collection contains only two-symbol cards. There are exactly six, so it must contain all of them.

**Hold back, then offer.** First eliminate singleton cards using P7. If triples remain troublesome, ask what happens to containment when every picture is replaced by the pictures missing from its card. Prove that step rather than merely naming complement.

# Sources extensions and verification

## Mathematical source and limits

Richard P. Stanley, MIT 18.318 *Topics in Algebraic Combinatorics*, Spring 2006, section 4, The Sperner property, printed pp. 17-25. Chain and antichain definitions appear on pp. 18-19; Proposition 4.4 and its proof on pp. 20-21 explain the chain-partition bound; Corollary 4.8 and Lubell's proof supply broader adult context. Source body verified October 3, 2026: read the MIT lecture notes.

The exact finite chain partitions, start-constrained optima, and completeness arguments in this guide were independently checked against the packets. The general Sperner theorem gives the middle-level binomial coefficient, but three small cases do not prove it. This hour does not require its algebraic proof or any factorial notation. This is subset containment, not the triangulation-labeling result also called Sperner's lemma.

Natasha Rozhkovskaya, *Math Circles for Elementary School Students*, Introduction: Berkeley 2009 (local EPUB, Part 1). The introduction describes individual adult assistance and active parent participation. Our short launch, hint timing, and hour menu are proposed adaptations, not reported classroom results.

## Extensions with clear answers

Complement every card in a legal collection relative to the active symbol set. The new collection is legal: X is contained in Y exactly when the complement of Y is contained in the complement of X. This explains the paired singleton/triple obstructions in upper P8 and why all singleton and all pair triples are companions in the three-symbol deck.

Ask whether removing two cards from the three-symbol deck always leaves a legal triple. No: remove A and BC. These hit both of the only size-three collections, A/B/C and AB/AC/BC, so a triple is impossible. After this specific removal, B/C achieves size two. Thus the remaining maximum is exactly two.

For the four-symbol deck, removing any two-symbol card reduces the maximum to five: the unique six-card maximum is broken and the five other pair cards remain legal. Removing any non-pair card leaves all six pair cards intact. These are consequences of the proved uniqueness, not a new general theorem.

## Independent audit and pilot notes

An independent check script in the source package (facilitator-src/independent_checks.py) represents cards as sets and exhaustively enumerates families without importing the student code. It finds 6, 20, and 168 legal collections including the zero-card family for two, three, and four symbols, and maxima 2,3,6. It checks all constrained starts, the nine two-card families, each removal in K-1 P8, and exact one-time coverage of every chain certificate. Output is independent-checks.json.

No pilot has occurred. Record whether children confuse overlap with containment, whether the empty card is treated correctly, and whether the row certificate arises naturally or requires more concrete work. Proposed timing and hints remain untested. The guidance revision adds a two-card nested-row example before each band's first row-arrangement task; the numbered investigations are retained.

# Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

### Use and return choices

Keep one adult at each table; reading and recording can be supplied while children rearrange cards. On another visit, K-1 can extend the fixed starts (P5) or make nested rows (P6), with P7-8 for children who can use a row certificate. Grades 2-3 can revisit small row partitions (P4-5), then the sixteen-card deck (P6-8). Grades 4-5 can keep the sixteen-card optimum and pursue row certificates (P4-6), forced-card maxima or all largest families (P7-8). Preserve the mathematical choices already in the packet.